

# WILLI WOLFF

Software Engineer · Technical Architect · Builder

Berlin, Germany · German & English

[Email](#) [LinkedIn](#) [GitHub](#)

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## PROFILE

Software engineer with 10+ years across enterprise (Deloitte), blockchain infrastructure (IOG/Cardano), mobile, and IoT. Currently technical architect for a Byzantine-resistant pub/sub network at IOG. Founder DNA, deep in distributed systems and secure software — curious by default, focused when it counts.

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## EXPERIENCE

### Technical Architect – Byzantine-Resistant PubSub Network

Feb 2026 – Present

*Input Output Global (IOG) · Contract*

Leading research and development of a Byzantine-resistant publish/subscribe network, formally verifying a research proposal and building a prototype.

- **Technical Leadership:** Lead a team of 4 engineers and researchers (plus myself) designing a Byzantine-resistant pub/sub overlay network
- **Formal Verification:** Formally verifying a research proposal and building a working prototype
- **Protocol Security:** Reevaluating peer-sampling protocols that come with specific security properties

### Technical Architect – Ouroboros Leios

Nov 2024 – Feb 2026

*Input Output Global (IOG) · Contract*

Led research and development of Ouroboros Leios, a high-throughput overlay protocol for Cardano blockchain, coordinating cross-functional team and driving toward mainnet implementation.

- **Technical Leadership & Team Coordination:** Direct team of 8 researchers and engineers across formal methods, protocol design, and prototype development
- **Defining Direction:** Translated open research into executable roadmap—authored [CIP-0164](#) (Cardano Improvement Proposal, industry RFC comparable to Bitcoin's BIPs or IETF standards)
- **Architecture & Strategy:** Designed simulation-driven approach validating protocol behavior before mainnet; created decision log tracking architectural choices and rationale
- **Cross-Org Coordination:** Bridge formal methods researchers, protocol engineers, and product teams—navigating complex technical trade-offs
- **Communication:** Host monthly public demos and Q&A sessions (5k+ average views); translate complex research for diverse audiences
- **Workshop Facilitation:** Organized week-long on-site workshop that reduced open research questions and aligned team on implementation strategy

### Technical Architect – Lace Wallet

Nov 2022 – Nov 2024

*Input Output Global (IOG) · Contract*

Led technical architecture and backend infrastructure for Lace, a next-generation Cardano wallet, delivering significant performance improvements and new revenue-generating features.

- **Backend Re-Architecture:** Re-engineered from poll-based to event-driven architecture—45% latency reduction, 60% + infrastructure cost savings

- **Custom Chain Indexer:** Built indexer with bloom filters and credential-based indexing; 2,000+ tx/s ingest capacity, sub-second wallet sync
- **Industry Standards:** Authored [CIP proposal](#) for wallet indexing standards adopted in ecosystem
- **Product Leadership:** Led 7-person cross-functional team (3 FE, 2 BE, 1 QA, 1 PM) delivering MetaDex swap feature
- **Shared Tooling:** Built transaction construction libraries and event-driven patterns adopted across multiple product teams
- **Technical Innovation:** Developed background web worker sync approach later adopted by team as standard pattern

## Solutions Architect

Nov 2021 – Nov 2022

*Input Output Global (IOG) · Contract*

Designed blockchain solutions and educated enterprise partners on Cardano's technical architecture.

- **Smart Contract Architecture:** Designed complete Plutus stack for collateralized debt position (CDP) platform
- **Technical Education:** Conducted workshops for 100+ Amazon engineers on eUTxO model, smart contracts, and dApp patterns
- **Knowledge Transfer:** Created training materials and hands-on exercises for blockchain education

## Senior Software Engineer

Feb 2020 – Nov 2021

*Mobility Data Lab GmbH, Berlin*

Built IoT and data processing systems for connected vehicle platform.

- **Fleet Management:** Delivered complete OTA fleet-update pipeline for connected vehicle hardware
- **IoT Platform:** Built OTA update mechanism (Rust embedded, Vert.x backend) with rollback capabilities
- **Analytics:** Designed real-time data pipeline using ELK stack for Miles and Share Now car-sharing services
- **Pilot Launch:** Successfully launched with 20+ vehicles

## Software Engineer → Lead Engineer & Senior Software Engineer

Sep 2016 – Nov 2019

*Deloitte, San Francisco*

Joined as Software Engineer and was promoted to lead a large-scale legacy system migration for a major German insurance company.

- **Migration Toolchain:** Redesigned Cobol-to-Java architecture for functional language (ISA); 3-stage pipeline: parser → AST → Java codegen, plus runtime emulation layer preserving original ISA/IDM behavior
- **Business Impact:** Saved client €6M annually in licensing costs
- **Team Growth:** Promoted to Lead within 24 months; scaled team from 3 to 20 engineers
- **Global Coordination:** Coordinated with offshore teams in India; established testing standards adopted organization-wide

## Co-Founder & CTO

Feb 2015 – Sep 2016

*CityXcape Inc., San Francisco*

Technical co-founder of social discovery platform for exploring cities through community-driven recommendations.

- **Product Development:** Built native iOS app (Swift) and Django/Rails backend from concept to App Store launch
- **Fundraising:** Raised €150k in seed funding from angel investors
- **Team Leadership:** Led technical team of 2-3 contract developers; defined product vision and roadmap

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## PUBLISHED WORK

Canonical architecture and design specification for Cardano’s high-throughput overlay protocol—an open industry RFC comparable to Bitcoin’s BIPs and IETF standards. Read it at [CIP-0164](#).

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## PROJECTS

### Ouroboros Leios – Protocol Architecture & CIP

2024 – Present

*Cardano Research*

Co-authored protocol specification and architecture for Cardano’s next-generation throughput layer.

- **CIP Authorship:** Principal author of [CIP-0164](#)—the canonical architecture design document guiding implementation teams
- **Protocol Specification:** Translated academic research into implementable specification with concrete parameters and trade-offs
- **Technical Documentation:** Created comprehensive [documentation site](#) covering protocol mechanics, simulation results, and design rationale

### Dump it! – AI-Enhanced Note Taking

2023 – 2025

*iOS, Android, Web*

Cross-platform application with end-to-end encryption and sophisticated AI processing pipeline.

- **Full-Stack:** Native iOS (Swift/SwiftUI), Android (Kotlin/Compose), and React web app with feature parity
- **E2E Encryption:** Diffie-Hellman key exchange + AES-256-GCM + iOS Secure Enclave—zero backend access to user data
- **AI Pipeline:** Two-phase LLM processing (Claude for intent, pgvector for semantic search, Whisper for speech-to-text)
- **Architecture:** Event-sourced with SSE streaming; offline-first with persistent queue sync
- **Traction:** 5k+ notes, 200+ users, 70% weekly retention; shipped to App Store and Google Play

### Pagebase – Freelance Development Agency

2024 – Present

*Founder*

Berlin-based agency building Supabase/PostgreSQL backends, web and mobile apps, and AI automation.

- **Approach:** Open-source first, with a focus on clean architecture and avoiding vendor lock-in
- **Stack:** Supabase, PostgreSQL, React/React Native, edge functions, CI/CD

### Drip Dropz – NFT Marketplace

2022 – 2024

*Cardano*

Production NFT trading platform built from scratch.

- **Frontend:** React/TypeScript SPA with CIP-30 wallet integration and background web worker sync
- **Smart Contracts:** Plutus integration for bidding, selling, swapping, and order management
- **Launch:** 10k+ NFT listings; first revenue within 48 hours
- **DevOps:** TDD practices with Jest; CI/CD pipeline for reliable deployments

### Berlin Pool – Stakepool Operations

2021 – 2024

*Cardano Infrastructure*

Operated Cardano consensus nodes contributing to network decentralization.

- **Node Operations:** Produced 2,000+ mainnet blocks; maintained 99.9% uptime; 15M+ ADA delegation
- **Security:** Air-gapped cold key storage; secure key rotation procedures

- **Infrastructure:** Automated deployment with Docker, Ansible, Terraform; Prometheus/Grafana monitoring

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## TECHNOLOGIES

### Languages

TypeScript (expert), Rust, Swift, Java/Kotlin, Python, Haskell/  
Plutus

### Security

E2E encryption, Diffie-Hellman, AES-256, Secure Enclave/TEE,  
Key management

### Web & Mobile

React, React Native, NestJS, Swift/iOS, Kotlin/Android

### Blockchain

Cardano (eUTxO), Plutus, CIP standards, Consensus protocols,  
Wallet standards

### Infrastructure

Distributed systems, Event-driven, PostgreSQL, Redis, gRPC,  
Microservices

### AI/ML

LLM integration (Claude, GPT-4), Vector search, Embeddings,  
Whisper

### DevOps

Docker, Terraform, Ansible, GitHub Actions, CI/CD, AWS,  
Prometheus/Grafana

### Leadership

Team building, Technical strategy, Architecture docs, Workshop  
facilitation

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## EDUCATION

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| <b>M.A. International Business</b> · Hult International Business School, San Francisco | 2014 – 2015 |
| <b>B.Sc. Computer Science</b> · DHBW Stuttgart, Germany                                | 2011 – 2014 |